

07/20/2026

Finished Work:

1. I have finished the selection of emission signals from the whole dataset. The selection process is a two-step clustering. First, a K-Means clustering is done on the **input voltage data**. For that, I set two (2) clusters, so it is 2-Means. Next, a GMM clustering is done by using two (2) Gaussians. Finally, results of the two clustering methods are compared. Data points with the same cluster indices are labeled with that common index. Data points with different cluster indices from different methods are labeled -1 and masked for non-use. In other words, data points with -1 labels will be ignored in analysis.
2. After the labeling procedure of the **input voltage data**, data indices are used in the discrimination of the **impedance measurement data** where it is low or high. Points with the low label are classified as emission data.
3. The user is allowed to select the emission data of interest in the data processing code. Either a single emission section can be chosen or all of the emission data can be chosen. In case of selection of all emission data, all the selection regions, their averages, and their standard deviations (as error) are reported.

Observations:

1. **Under the assumption of identical/similar standard deviations throughout the data**, a histogram of the collapsed data standard deviations shows a resemblance to a Gaussian distribution for the lowest capacitance-measured data set, way more than the other data sets, which are grouped around a higher capacitance value. Similar collapsed standard deviations of those data sets show a clear resemblance to a heavy-tailed distribution (i.e., Lognormal distribution). This is shown in the following figures;



